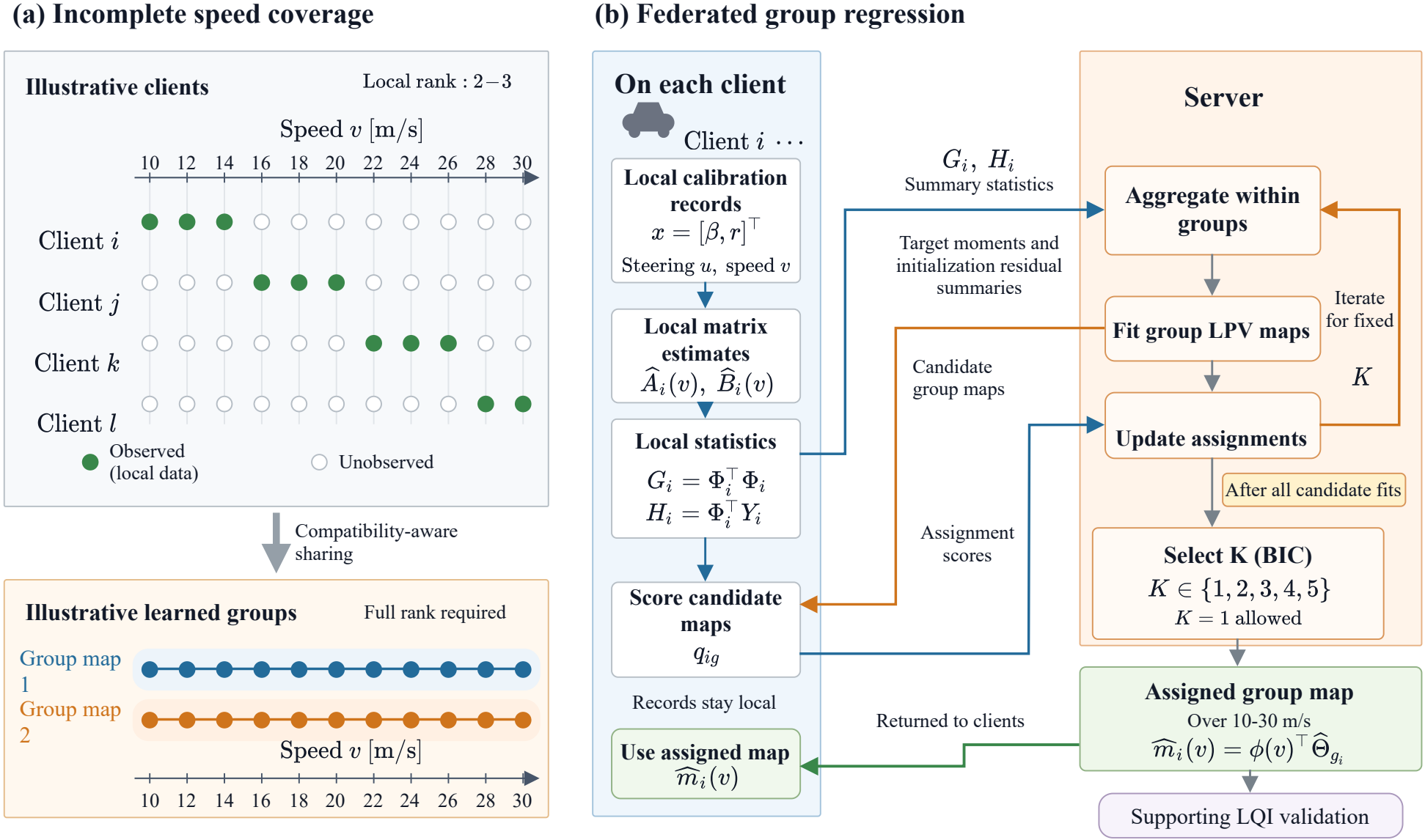

(a) Incomplete speed coverage
Illustrative clients
Local rank : 2 – 3
Speed v [m/s]
10 12 14 16 18 20 22 24 26 28 30
Client i
Client j
Client k
Client l
Observed (local data)
Unobserved
Compatibility-aware sharing
Illustrative learned groups
Full rank required
Group map 1
Group map 2
Speed v [m/s]
10 12 14 16 18 20 22 24 26 28 30
(b) Federated group regression
On each client
Client i ⋯
Local calibration records
$x = [\beta, r]^\top$
Steering u, speed v
Local matrix estimates
$\widehat{A}_i(v)$, $\widehat{B}_i(v)$
Local statistics
$G_i = \Phi_i^\top \Phi_i$
$H_i = \Phi_i^\top Y_i$
Score candidate maps
q_{ig}
Records stay local
Use assigned map
$\widehat{m}_i(v)$
G_i, H_i
Summary statistics
Target moments and initialization residual summaries
Candidate group maps
Assignment scores
Returned to clients
Server
Aggregate within groups
Fit group LPV maps
Iterate for fixed K
Update assignments
After all candidate fits
Select K (BIC)
$K \in \{1, 2, 3, 4, 5\}$
$K = 1$ allowed
Assigned group map
Over 10-30 m/s
$\widehat{m}_i(v) = \phi(v)^\top \widehat{\Theta}_{g_i}$
Supporting LQI validation